Sundaram Tripathi

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https://github.com/thefinisher1

Summary

Electrical Engineering student experience in Drone Systems, Embedded Controllers, and Mechatronics. Instrested to design and integrate complex hardware-software systems, including autonomous drones and assistive medical technology. Focused on leveraging electrical fundamentals and automation to solve real-world challenges

Education

Bachelor of Technology in Electrical Engineering — Sept 2023 – May2028

- G.L. Bajaj Institute of Technology and Management, Greater Noida, UP
- CGPA: 4.32
- 2024-25: Failed

Technical Skills

- Microcontrollers: Arduino, ESP32 / ESP8266, Raspberry Pi Pico, Arduino Pro Micro
- Hardware & Systems : Embedded Systems, Microcontrollers, Laser Cutting Machine Operation, CNC Machine Operation, 3d printing, System Integration
- Electrical Core : Electrical Engineering Fundamentals, System Design, Power Electronics, Circuit Design
- Software & Tools: Arduino IDE, VS code, Auto Cad 2025, KI Cad, Fusion 360, Esp-IDF, Cura, Simply3d, Mission planner, Beta flight configurator

Projects

Intelligent Disaster Response Drone System - *Embedded Systems, Automation*

- Engineered an automated aerial system designed to reduce emergency response time and enhance situational awareness during natural and man-made disasters.
- Integrated embedded controllers with real-time monitoring systems to perform precision aerial surveillance and identify impacted zones or stranded individuals.
- Developed a scalable system architecture focusing on flight stability and rapid deployment for disaster-prone environments.
- Applied a multidisciplinary approach combining electrical engineering fundamentals and controller programming to deliver a reliable search and rescue solution.

Affordable Smart Prosthetic Hand - *Mechatronics, Embedded Systems, Robotics*

- Developed a cost-effective and accessible alternative to traditional prosthetics, focusing on balancing performance with affordability.
- Synthesized mechanical structural modelling with electronic control systems to simulate natural gripping and release actions via actuator integration.
- Implemented controller-based motion execution while prioritizing ergonomics and simplicity for daily practical usability.
- Demonstrated innovation-driven problem solving by making complex assistive technology accessible to a wider demographic.

Interests & Technical Focus

- Technical Interests : Embedded Systems, Industrial Automation, Robotics
- Focus Areas : System Efficiency, IoT-enabled Electrical Systems, Hardware-Software Interfacing
- Activities : Actively exploring advancements in IOT through hands-on prototyping.